Pricing Azerbaijani real estate on bina.az

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Task A

Regression: predict the sale price in AZN

Task B

Classification: premium vs standard, split at the training median (210,000 AZN)

Kaggle bina.az sale dump: 64,247 clean listings, 91 features, split 70 / 15 / 15

Cleaning removes leakage and repeated scrapes

~36k

extra rows from repeated scrapes of one URL; keep the latest only

2

leakage columns removed: unit_price and total_price restate the target

91

features after encoding; levels under 100 train listings become other

- Outliers by fixed bounds: price 5,000 to 10,000,000 AZN, area 10 to 100,000 m², rooms 1 to 15
- Units: sot converted to m², floor '7 / 9' split into floor and floor_total
- Split first; medians, scaler, levels and tier threshold learned on train only

Decision tree: one sorted pass finds every split

Impurity

Gini, entropy for classes; variance (MSE) for regression. Gain = I(parent) minus weighted child impurity

Split search

Sort each feature once; cumulative counts (or sums of y and y^2) score every cut at once; threshold = midpoint

- Stopping: max_depth, min_samples_split, min_samples_leaf, min_impurity_decrease (each unit-tested)
- Task A chosen: depth 8, min leaf 50, validation R^2 0.579
- Task B chosen: Gini, depth 14, min leaf 1, validation F1 0.919

Pegasos SVM: sub-gradient descent on the hinge primal

$J(w,b) = \lambda/2\ \|w\|^2 + \text{mean of } \max(0, 1 - y(w^T x + b))$

Update: $w \leftarrow (1 - \eta\lambda)\, w + (\eta/k) \sum y \cdot x$ over violators in a batch of $k = 64$

- $\eta = 1/(\lambda t)$, then projection onto $\|w\| \le 1/\sqrt{\lambda}$; average the second half of the iterates
- Exposed: λ, training length (epochs), learning-rate schedule; reports margin $2/\|w\|$ and support vectors
- $\lambda = 10^{-3}$: margin 0.78, 16,906 support vectors (38% of train)
- Non-linear: random Fourier features ($D = 300$) approximate the RBF kernel, same solver

Setup: matched baselines, validation-only tuning

Seed 42

one command, python -m src.run_all, reproduces every number

Validation

all hyperparameters chosen there; test used once

Uncertainty

200 bootstrap resamples plus 5-fold CV mean ± std

- Metrics: RMSE, MAE, R^2 (A); accuracy, precision, recall, F1, ROC-AUC, PR-AUC (B)
- Baselines: scikit-learn trees (same depth and leaf), SGDClassifier(hinge, alpha = λ), SVC(rbf, C = $1/(\lambda n)$)
- Scaling in a Pipeline; SVC trained on an 8,000-row subsample

Results: our models match scikit-learn

Model (test set)	Task A R^2	Task B F1	ROC-AUC
Our tree	0.533	0.922	0.924
sklearn tree	0.533	0.919	0.920
Our linear Pegasos SVM	n/a	0.850	0.931
sklearn SGD hinge	n/a	0.850	0.930
Our RFF RBF SVM	n/a	0.849	0.932
sklearn SVC (8k rows)	n/a	0.855	0.936

- Task A RMSE 342,525 (ours) vs 342,515 (sklearn); 99.1% identical Task B predictions
- Tree: best F1. SVMs: best ranking by AUC. Pegasos fits as fast as SGD and predicts faster

Where the models fail, and the limits

61%

median error on land plots (flats: 14 to 15%)

50% vs 17%

error in sparse districts vs elsewhere

24% vs 4%

Task B errors within ±15% of 210k vs elsewhere

- Houses 27%, commercial objects 47%; luxury quartile MAE 268k AZN drives RMSE
- Limits: R^2 only 0.53, SVC on 8,000 rows, descriptions unused, asking prices, tree 5 to 18 times slower

Contributions and next steps

Who did what

- All: data cleaning and EDA
- Araz, Elsaf: decision tree and experiments
- Huseynali, Joshgun: Pegasos SVM, RFF, experiments
- Huseynali, Araz: benchmark, error analysis, report

Next

- Random forest or boosting from our own trees
- Text features from listing descriptions
- Geographic clustering to replace sparse districts